Fleet-Wide Condition Monitoring via Shared Nominal Representations and Local Radius Calibration

Enabling OEM-Centralized, Fleet-Deployed CBM

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Abstract

Objectives: Condition monitoring is normally deployed by collecting data from each asset, fitting a detector to it, and engineering its thresholds, so effort grows with fleet size. This work asks whether representation learning can be decoupled from deployment: one representation shared fleet-wide, each asset identifying only its own local operational radius at commissioning.

Methods: For each asset family, a canonical nominal representation, a convolutional autoencoder, is trained once offline on simulated healthy signals and transferred unchanged to every deployed unit. Each unit estimates its operational radius as an empirical residual quantile from a short unlabeled nominal stream, stopped automatically by a convergence criterion. Bearings serve as a validation vehicle: the Case Western Reserve University (CWRU) dataset across four operating conditions, and four run-to-failure bearings from the NASA Intelligent Maintenance Systems (IMS) dataset, benchmarked against threshold transfer, oracle calibration, and asset-specific detectors.

Findings: Transferring the absolute simulator threshold flagged every healthy CWRU window as anomalous: the representation transfers, its error scale does not. Commissioning recovered radii within 0.83% of the complete-data oracle after 55 windows on average, yielding a healthy false-alarm rate of 2.60%, 99.57% ball-fault detection and 100% inner- and outer-race detection; all 20 repeated CWRU realizations converged. The four IMS bearings converged to distinct radii and departed persistently from them on average 45.4 hours before the documented end of run.

Novelty: The contribution is a fleet-deployment methodology rather than a new anomaly-detection architecture: the transferable artefact is a validated shared nominal representation, and the only asset-specific quantity estimated at deployment is a scalar operational radius.

Practical Applications: A component manufacturer or analytics provider can validate, version, and maintain one representation per asset family and distribute it unchanged, so commissioning needs no asset-specific data, training, or threshold engineering, only a short unsupervised calibration suited to edge deployment.

Keywords: condition-based maintenance (CBM), transport asset management, fleet-wide deployment, asset commissioning, bearing condition monitoring, OEM deployment, shared nominal representation, simulation-to-field transfer, edge AI, run-to-failure validation

INTRODUCTION

Modern transportation systems increasingly rely on condition-monitoring technologies to supervise large populations of operational assets, including rolling-element bearings, axleboxes, traction motors, gearboxes, pumps, actuators, vehicles, and spatially indexed infrastructure segments (Jardine et al. 2006; Zhao et al. 2019). Railway axlebox bearings alone are monitored across thousands of independently operating vehicles, where undetected defects can propagate to derailment-level failures (Entezami et al. 2020). Although machine-learning research has produced increasingly capable anomaly-detection models (Chandola et al. 2009; Zhao et al. 2019), the deployment problem remains largely unresolved: a detector that performs well for one asset under one calibrated operating condition does not automatically scale to hundreds or thousands of independently commissioned units.

Fleet growth therefore scales not only the number of monitored assets but also the engineering effort required to train, validate, version, and maintain asset-specific models, and at rollout several constraints arise simultaneously. Representative fault examples are rarely available, and waiting months or years for failures to accumulate is operationally unacceptable. Installation factors such as sensor coupling, operating point, mounting, and environmental noise alter the numerical scale of anomaly scores between assets (Randall and Antoni 2011): axlebox bearings of nominally identical rolling stock are exposed to different track quality, wheel wear, load distribution, and wayside versus on-board sensor placement, so that a detector calibrated on one vehicle rarely transfers its absolute decision threshold to another (Entezami et al. 2020).

The practical question is therefore not how an anomalous condition can be detected, but rather:

> **Q1.** *How can a single nominal representation be deployed across a heterogeneous fleet while allowing every physical asset to identify only its own local operational neighbourhood, without per-asset representation learning or manual threshold engineering?*

This paper addresses Q1 by separating representation engineering from deployment calibration. A canonical nominal representation is learned once, offline, for a compatible asset family and deployed unchanged across the corresponding fleet, remaining fixed throughout the asset lifetime. During commissioning, each physical asset identifies only its own operational radius from a short stream of unlabeled nominal observations. This defines an OEM-centralized deployment model: the representation is engineered, validated, versioned, and maintained centrally, for example by the original equipment manufacturer (OEM), while fleet expansion requires only distributing that representation and estimating one local radius per asset.

The transferable component is therefore not an anomaly threshold but the shared nominal representation itself, whose absolute residual scale depends on simulator, preprocessing, and operating conditions and does not transfer. Deployment is thus a local inverse problem: it does not learn a new representation, but estimates an asset-specific property of nominal operation under a fixed shared representation and calibration protocol, following the formulations introduced in our previous work (Di Santi 2026a,b).

This formulation also changes the relevant experimental comparison. The appropriate reference is not another globally shared detector but an operationally unrealistic upper bound in which every physical asset has its complete nominal history available for independent representation learning and calibration. Evaluating Q1 therefore requires answering a second question:

> **Q2.** *What is the operational cost of replacing independently engineered asset-specific representations with one centrally maintained shared representation and lightweight local commissioning?*

Q2 is evaluated in two complementary scenarios. The first uses the CWRU (Smith and Randall 2015) benchmark to quantify simulation-to-field transfer under controlled operating-condition variability, comparing the proposed methodology with ideal asset-specific representations trained and calibrated on the complete healthy dataset of each condition. The second uses the NASA IMS run-to-failure dataset to evaluate the complete deployment protocol on independently monitored physical assets undergoing naturally evolving degradation. In both, the proposed method uses no field observations for representation learning and estimates the local operational radius through convergence-based commissioning, requiring only tens of nominal observations. The results show that the cost of scalable deployment is small: detection performance remains close to that of independently engineered asset-specific detectors, with no field retraining at any point.

The contribution of this paper is therefore not a new anomaly-detection architecture but a deployment methodology for fleet-scale condition monitoring. Although a Conv1D autoencoder is used as the experimental representation, the deployment principle is representation-agnostic. Rolling-element bearings serve as a validation vehicle rather than as the target application: they were selected because public run-to-failure data with documented ground truth exists for them, which permits the deployment claim to be tested against an observable outcome. The principal contributions are:

1. an OEM-centralized fleet-deployment architecture in which one shared canonical nominal representation is engineered, validated, and maintained centrally and transferred unchanged across heterogeneous assets, with no field-data retraining and no per-asset representation learning after installation;
2. an online commissioning procedure that estimates each asset's local operational radius from a short stream of unlabeled observations, together with a convergence criterion that determines automatically when to stop and freeze it;
3. experimental evidence that independently monitored physical assets identify substantially different local operational radii while sharing the same representation, target nominal coverage, and convergence criterion; and
4. an evaluation against ideal asset-specific detectors trained and calibrated with complete healthy datasets.

The remainder of the paper is organized as follows. Section 2 reviews condition monitoring, one-class anomaly detection, adaptive calibration, and fleet deployment. Section 3 introduces the deployment methodology. Section 4 presents the experimental protocol. Section 5 reports the deployment results. Section **??** closes with a discussion of industrial implications, limitations, and future work.

RELATED WORK

Condition-based maintenance (CBM) has become a central component of modern transportation asset management, where condition monitoring supports the operation of large fleets of heterogeneous assets (Jardine et al. 2006; Chandola et al. 2009; Zhao et al. 2019). Considerable progress has been made in improving anomaly-detection algorithms; however, comparatively little attention has been paid to how these detectors are engineered, commissioned, and maintained at fleet scale once thousands of independent physical assets must be deployed.

Deployed practice, however, is largely not model-based. For rotating machinery the established approach combines physically interpretable features with severity criteria standardized in ISO 20816, which superseded ISO 10816 and ISO 7919: an absolute criterion, in which tabulated zone boundaries are applied without per-unit calibration, and a change criterion referenced to a baseline established for each installed machine. Envelope demodulation complements this by attributing spectral peaks to kinematic defect frequencies, yielding fault type rather than fault presence. This matters for the present work in two ways. First, absolute thresholds demonstrably do transfer in that setting, so the non-transferability reported here is a property of learned reconstruction residuals, not a general claim. Second, the kinematic route still requires per-asset engineering, since bearing geometry, shaft speed, and monitored bands must be configured for every installed unit; the effort is human rather than computational, but it still scales with fleet size. The deployment problem addressed here is therefore sharpest for asset families with no tabulated severity chart and no kinematic defect model.

Most machine-learning approaches formulate this problem as supervised fault classification, requiring representative fault examples during training. While effective on benchmark datasets, these methods assume that deployment conditions remain close to those observed during model development.

One-class methods, including one-class SVM (Schölkopf et al. 1999), neural-network novelty detection (Bishop 1994), and autoencoder-based anomaly detection (Hinton and Salakhutdinov 2006; Sakurada and Yairi 2014), reduce the dependence on fault labels by learning only nominal behaviour. More recent representation-learning approaches further improve anomaly detection by constructing latent spaces in which healthy operation occupies a compact region while degraded behaviour becomes more easily separable.

Adaptive thresholds have been widely used to compensate for changing operating conditions. However, unrestricted adaptation risks absorbing progressive degradation into the nominal baseline. This motivates separating calibration from monitoring. In the proposed methodology, adaptation is limited to a finite cold-start phase during which each asset estimates its local operational radius. Once convergence is achieved, the shared representation remains fixed and subsequent departures become diagnostic evidence rather than input for further adaptation.

The proposed deployment strategy is also related to invariant representation learning (Bronstein et al. 2021) and anomaly detection under distribution shift (Carvalho et al. 2023), both of which seek representations that are insensitive to nuisance variability.

A related and rapidly growing literature addresses cross-condition and cross-machine transfer through domain adaptation, typically aligning source and target feature distributions via maximum mean discrepancy, adversarial domain classifiers, or pseudo-labeled fine-tuning (Nguyen et al. 2025; Ning et al. 2026). Closest to the present work, Wang et al. (2026) propose a two-stage unsupervised transfer framework for fleets: an encoder–decoder is pre-trained on one source system for which a long operational history exists, and its architecture and parameters then initialize a *dedicated* model for each target system of the same fleet, validated in part on traction systems of a fleet of trains. These methods substantially reduce the need for labeled target-domain faults and have demonstrated strong cross-condition accuracy. However, they still require target-domain observations during an explicit adaptation stage executed independently for each new operating condition or physical unit. This reintroduces, at a smaller scale, the same per-asset engineering burden that fleet-wide deployment seeks to eliminate: every newly installed asset still triggers a training procedure, even when it requires no fault labels.

The distinction is therefore not between supervised and unsupervised deployment, nor between labeled and unlabeled adaptation, but between architectures in which each new asset triggers engineering work and one in which deployment consists only of estimating a local operational radius while reusing an already validated shared representation. Existing transfer approaches transfer model parameters and then adapt them per asset; here the transferable artefact is the validated nominal representation itself, and the only asset-specific quantity estimated at deployment is that radius.

This viewpoint is also consistent with the inverse-problem and quotient-space formulations introduced in our previous work (Di Santi 2026b,a), in which diagnosis is posed as the recovery of a canonical nominal representation from observations. The present paper addresses the engineering realization of that framework for scalable fleet deployment rather than its mathematical formulation.

METHOD

This section formalizes the proposed deployment methodology. The central idea is to separate representation engineering from deployment commissioning. A single nominal representation is engineered once and shared across the fleet, whereas each deployed asset identifies only its own local operational radius during commissioning.

Deployment architecture: shared representation, local operational radius

Let $x_{k,t} \in \mathcal{X}$ denote an observation acquired from operational asset k at time t. In the present experiments, $x_{k,t}$ corresponds to a fixed-length vibration recording.

A shared nominal representation, trained once offline, maps each observation to a scalar residual score

$$r_{k,t} = r(x_{k,t}). \tag{1}$$

During commissioning, each deployed asset estimates its own local operational radius, denoted by $\tau_k^\star$, from a short sequence of unlabeled nominal observations. This radius characterizes the local neighbourhood of the nominal manifold represented by the shared model. Once commissioning converges, the operational radius is frozen and remains fixed throughout subsequent monitoring.

An alarm is therefore generated whenever

$$r_{k,t} > \tau_k^\star. \tag{2}$$

The deployment methodology follows three principles:

1. one centrally engineered, validated, versioned, and governed nominal representation;
2. one lightweight commissioning procedure performed independently by each deployed asset;

3. no representation retraining, fine-tuning, or parameter adaptation after deployment.

This separation directly matches the OEM-centered deployment model introduced in Section 1. The shared representation is engineered, validated, versioned, and maintained centrally by whichever organization owns the diagnostic model, for example the component OEM, a third-party analytics provider, or the fleet operator itself. Each deployed asset performs only a lightweight local commissioning procedure in which its own operational radius is identified automatically from nominal observations.

Consequently, fleet growth no longer requires engineering additional representation models. Deploying a new physical asset consists only of distributing the already validated shared representation and estimating one local operational radius for that asset.

The transferable engineering artefact is therefore the nominal representation itself rather than an absolute decision threshold. Different assets identify different operational radii while sharing exactly the same representation, the same target nominal coverage, and the same commissioning procedure.

Structural and coverage properties. Two properties support this architecture. First, any residual constant on nuisance-induced equivalence classes factors through the diagnostic quotient, so a single fleet-wide representation is structurally admissible. Second, if the commissioning residuals and a future nominal residual are exchangeable, a radius taken as the k-th order statistic with $k = \lceil (1-\alpha)(n+1) \rceil$ satisfies $P(r_{n+1} \leq \tau) \geq 1-\alpha$ without any assumption on the residual distribution; attaining the target coverage requires $n \geq 1/\alpha - 1$, or 49 windows at $\alpha = 0.02$, which is consistent with the commissioning durations reported in Section 5. The convergence criterion makes n data-dependent, so the bound holds approximately rather than exactly. The full functional-analytic and topological development is given in our companion work (Di Santi 2026a,b).

EXPERIMENTAL DESIGN

The experimental protocol evaluates the proposed deployment methodology rather than the underlying representation-learning architecture. Across the five experimental stages, we measure:

- number of observations required for operational-radius convergence;
- variability of $\tau_k^\star$ across assets and operating regimes;
- post-commissioning healthy false-alarm rate;
- post-commissioning fault and persistent-departure detection;
- robustness to commissioning-sample variability and arrival order;
- agreement between online and complete-data calibration; and
- deployment cost avoided by eliminating per-asset representation learning.

Stage 1 — Construction of the shared nominal representation. A physics-motivated nominal simulator generates healthy bearing signals using 1200-sample windows, a 30 Hz shaft frequency, and bounded variation in amplitude, phase, speed, offset, and additive noise. The shared Conv1D autoencoder is trained once on 4,000 simulated healthy windows, with an additional 1,000 nominal windows reserved for validation. Fault realizations are generated only for synthetic evaluation and are never used during representation learning.

The synthetic environment is used exclusively to construct the fleet-wide canonical nominal representation prior to deployment. Quantitative synthetic validation of the quotient-space formulation, including the convergence of independently commissioned synthetic assets toward stable asset-specific operational radii under the shared representation, is reported in the companion theoretical work (Di Santi 2026a,b). Since the objective of the present paper is to evaluate fleet deployment rather than simulation fidelity, the quantitative results reported here focus exclusively on transfer to the independent CWRU and NASA IMS benchmarks.

Stage 2 — Direct simulation-to-field deployment on CWRU. The nominal representation trained in Stage 1 is transferred unchanged to the Case Western Reserve University (CWRU) Drive-End dataset. Following common practice for this accelerometer channel, signals are sampled at 12 kHz, so that each 1,200-sample window corresponds to a 100 ms acquisition interval. No CWRU observation is used for representation learning, retraining, or fine-tuning.

Each operating condition is treated as an independent deployment environment. During commissioning, healthy observations are processed sequentially. Calibration begins after a minimum of $N_{\min}$ healthy windows and continues until either

(i) the operational radius satisfies the convergence criterion, or

(ii) a maximum calibration budget $N_{\max}$ is reached.

Once convergence is declared, the operational radius is frozen and all subsequent observations are evaluated using the unchanged shared representation.

This stage evaluates whether

(i) the learned representation transfers directly from simulation to field;

(ii) local empirical calibration compensates for the non-transferability of absolute reconstruction-error thresholds; and

(iii) the calibration duration can be determined automatically from the observed convergence behaviour rather than prescribed a priori.

The four CWRU operating conditions provide a controlled evaluation of score-scale heterogeneity and deployment calibration rather than four physically distinct fleet assets. Validation on independently monitored run-to-failure bearings is addressed in Stage 4 using the NASA IMS dataset.

Stage 3 — Reference comparison and calibration ablation. To quantify the cost of scalability, we construct an intentionally favorable but operationally non-scalable reference condition. For each CWRU operating condition, an independent Conv1D autoencoder and preprocessing transformation are trained using the complete healthy dataset. Each detector uses the empirical 98th percentile of its healthy reconstruction-error distribution as its decision boundary.

Using the unchanged simulator-trained representation, we compare three deployment strategies:

1. **Direct threshold transfer:** apply the absolute simulator threshold unchanged to each CWRU operating condition;
2. **Field oracle:** estimate the operational radius from the complete healthy dataset available for each operating condition, without modifying the shared representation;
3. **Proposed online calibration:** estimate the operational radius sequentially from healthy observations until the convergence criterion is satisfied or the maximum calibration budget $N_{\max}$ is exhausted.

This ablation separates representation transfer from calibration transfer. The representation is identical in all three deployment strategies; only the information available for estimating the local operational radius changes.

Stage 4 — Multi-unit run-to-failure validation on NASA IMS. A second nominal representation is constructed for the NASA IMS bearing family using simulated healthy observations matched to the dataset sampling and operating characteristics. The representation is trained once and transferred unchanged to each independently monitored bearing.

Each bearing estimates its operational radius from an early-life nominal segment using the same convergence-based calibration procedure adopted in the CWRU experiments. Once convergence is declared, the operational radius is frozen and monitoring continues without further adaptation.

This stage evaluates whether the proposed deployment methodology extends from heterogeneous operating conditions to physically distinct bearing units undergoing naturally evolving degradation.

We report (i) observations required for convergence;

(ii) variability of the converged operational radius across bearings;

(iii) healthy-period false-alarm rate;

(iv) first persistent departure after convergence; and

(v) alarm lead time relative to the documented end of the experiment.

Persistent departure is declared using a 10-recording persistence window. A departure begins at the first recording for which at least eight of the next ten recording-level residuals exceed the frozen operational radius. The declaration time is the final recording of that first qualifying window. Lead time is measured from the declaration time to the documented end of the experiment, using the 10-minute acquisition interval of the NASA IMS trajectory. Post-departure exceedance is evaluated from the detected departure onset to the end of the run.

Stage 5 — Edge deployment cost. We report model size, parameter count, CPU inference latency for one 1200-sample window, and local calibration memory requirements.

RESULTS

The results are organized around three complementary deployment questions. First, the CWRU experiments establish a favorable but operationally non-scalable upper reference in which an independent autoencoder, preprocessing transformation, and complete-data calibration are available for each operating condition. Second, they quantify whether one simulator-trained shared representation, combined only with convergence-based local commissioning, can recover a comparable diagnostic operating point without field-model retraining. Third, the NASA IMS run-to-failure experiment evaluates the complete deployment methodology on independently monitored physical assets undergoing naturally evolving degradation.

The CWRU evaluation considers four operating conditions corresponding to 1730, 1750, 1772, and 1797 RPM. Across these conditions, the experiments contain 1,414 healthy windows, 1,621 ball-fault windows, 1,621 inner-race-fault windows, and 2,846 outer-race-fault windows. All windows contain 1,200 samples and are subjected to the same evaluation protocol, while the representation and calibration information available to each deployment strategy differ according to the experiment.

The NASA IMS evaluation considers four independently monitored bearings from NASA IMS Test Rig Experiment 2, each represented by 984 chronologically ordered recordings. Unlike the controlled CWRU fault-window comparison, this experiment evaluates whether locally commissioned operational radii remain diagnostically meaningful throughout complete run-to-failure trajectories.

Ideal asset-specific reference

Table 1 establishes the practical upper reference obtained when independent autoencoders are trained and calibrated for each CWRU operating condition using the complete healthy dataset available for that condition. These experiments constitute an oracle reference rather than a deployable fleet architecture: they assume complete nominal data, an independently fitted preprocessing transformation, and separate representation learning and calibration for every operating environment.

Each asset-specific autoencoder uses the empirical 98th percentile of its own healthy reconstruction-error distribution as its decision boundary. Fault observations are never used for representation learning or threshold selection.

TABLE 1 Ideal asset-specific autoencoder performance on CWRU. Each autoencoder is trained and calibrated independently using the complete healthy dataset of the corresponding operating condition. FAR denotes healthy false-alarm rate and DR denotes fault detection rate.

RPM	Threshold	Healthy FAR (%)	Ball DR (%)	Inner-race DR (%)	Outer-race DR (%)
1730	0.6485	2.23	100.00	100.00	100.00
1750	0.7590	2.23	100.00	100.00	100.00
1772	0.7248	2.23	100.00	100.00	100.00
1797	0.4591	2.46	100.00	100.00	100.00
Mean	–	2.29 ± 0.12	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00

The asset-specific autoencoders achieve 100% detection for all evaluated fault families under every operating condition while reproducing the expected approximately 2% healthy false-alarm rate induced by empirical p98 calibration. This operating point serves as the reference against which the deployment cost of the proposed methodology is quantified in the following sections.

The absolute reconstruction-error threshold differs substantially across operating conditions, ranging from approximately 0.46 to 0.76. This variation is important for the deployment problem addressed in this paper: even when independently trained models produce identical diagnostic performance, their numerical score scales are not interchangeable.

The asset-specific results therefore define a favorable but operationally non-scalable reference. Each deployment environment receives its own preprocessing transformation, representation model, and complete-data calibration. The subsequent experiments quantify the operational cost of replacing this ideal deployment with one simulator-trained shared representation commissioned independently at each asset through local operational-radius estimation, without any field-model training.

Repeated-run stability of the asset-specific autoencoder

To assess the effect of stochastic optimization, the complete asset-specific training and evaluation procedure was repeated five times using independent random initializations. The architecture, preprocessing, datasets, training protocol, and empirical p98 calibration remained unchanged.

Across all operating conditions, healthy false-alarm rate and the detection rates for ball-, inner-race-, and outer-race faults remained unchanged across repetitions. Only the numerical reconstruction threshold varied, as summarized in Table 2.

TABLE 2 Repeated-run variability of the asset-specific reconstruction threshold over five independent random initialisations.

RPM	Reconstruction threshold
1730	0.649 ± 0.021
1750	0.759 ± 0.227
1772	0.725 ± 0.148
1797	0.459 ± 0.026

Although the detector was retrained five times on exactly the same healthy dataset, the resulting reconstruction thresholds differed because of stochastic optimization, while the corresponding diagnostic operating points remained unchanged.

This observation has an important implication for fleet deployment. Diagnostic performance is associated with the learned nominal representation rather than with a unique numerical residual scale. Different optimization trajectories produce diagnostically equivalent representations while assigning different absolute reconstruction values to them. Consequently, transferring an absolute threshold between models is generally not meaningful, whereas transferring the shared nominal representation and allowing each deployed asset to identify its own local operational radius is consistent with the observed behaviour.

Direct simulation-to-field deployment

The shared autoencoder is trained once using simulated healthy observations and transferred unchanged to all four CWRU operating conditions. No CWRU observation is used to retrain, fine-tune, or otherwise modify either the representation or its preprocessing transformation.

Three deployment conditions are evaluated:

1. **Direct threshold transfer**, in which the absolute empirical p98 residual threshold estimated from simulated nominal data is applied unchanged to each field operating condition;
2. **Field oracle**, in which the empirical p98 operational radius is computed retrospectively from the complete healthy field dataset of each operating condition, without retraining the shared representation;
3. **Proposed online calibration**, in which the local empirical p98 operational radius is estimated sequentially from a commissioning residual stream assumed nominal during initialization. Convergence is evaluated only after a minimum calibration size $N_{\min} = 40$, and calibration continues until the stability criterion is satisfied or the maximum budget $N_{\max} = 200$ is reached.

Once convergence is detected, the corresponding operational radius is frozen and used for subsequent monitoring. The commissioning duration is therefore determined independently for each operating condition rather than fixed in advance.

Table 3 reports the complete CWRU deployment results.

TABLE 3 Simulation-to-field deployment of the shared autoencoder on CWRU. FAR denotes healthy false-alarm rate, DR fault detection rate. Healthy FAR covers the complete healthy sequence, including commissioning observations, so that calibration strategies remain comparable; post-convergence FAR is reported in Table 5. Healthy windows per condition: 404 (1730, 1750), 403 (1772), 203 (1797). A dash indicates that sequential calibration does not apply.

RPM	Calibration	τ	FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Cal. windows
1730	Direct threshold transfer	0.1027	100.00	100.00	100.00	100.00	–
1730	Field oracle	0.3578	2.23	99.75	100.00	99.86	–
1730	Proposed online calibration	0.3520	3.71	99.75	100.00	100.00	49
1750	Direct threshold transfer	0.1027	100.00	100.00	100.00	100.00	–
1750	Field oracle	0.3434	2.23	100.00	100.00	100.00	–
1750	Proposed online calibration	0.3440	1.98	100.00	100.00	100.00	62
1772	Direct threshold transfer	0.1027	100.00	100.00	100.00	100.00	–
1772	Field oracle	0.3512	2.23	100.00	100.00	100.00	–
1772	Proposed online calibration	0.3550	1.24	100.00	100.00	100.00	58
1797	Direct threshold transfer	0.1027	100.00	100.00	100.00	100.00	–
1797	Field oracle	0.3886	2.46	98.03	100.00	100.00	–
1797	Proposed online calibration	0.3869	3.45	98.52	100.00	100.00	49

Direct transfer of the absolute simulator threshold fails under every operating condition. Although it produces an apparent 100% detection rate for all evaluated fault families, it simultaneously classifies every healthy field window as anomalous. The absolute residual scale learned in simulation therefore does not transfer directly to the field.

In contrast, local calibration restores an operationally meaningful decision boundary without modifying the shared representation. The complete-data field oracle reproduces the expected approximately 2% healthy false-alarm rate and achieves 99.45% average ball-fault detection, 100% inner-race detection, and 99.96% outer-race detection.

The proposed online procedure reaches a closely comparable operating point using only observations available during commissioning. All four operating conditions satisfy the convergence criterion before the maximum calibration budget is exhausted, requiring 49, 62, 58, and 49 windows for 1730, 1750, 1772, and 1797 RPM, respectively. The mean commissioning duration is therefore 54.5 ± 6.6 windows.

Across the four conditions, the proposed procedure obtains a mean healthy FAR of $2.60 \pm 1.18\%$, 99.57% average ball-fault detection, 100% inner-race detection, and 100% outer-race detection. These results remain close to the complete-data oracle despite requiring neither complete prior knowledge of the healthy field distribution nor independent representation learning for each operating condition.

For comparability between deployment strategies, Table 3 reports healthy FAR over the complete healthy sequence, including commissioning observations. The operationally relevant

FAR evaluated only after convergence is reported separately in Table 5.

The central result is not that the online procedure outperforms the complete-data oracle. Rather, it recovers a comparable operating point through a short, automatically terminated commissioning phase while retaining one fixed fleet-wide representation and performing no field-model training.

Cost of scalable deployment

Table 4 compares the proposed shared deployment with an ideal but operationally non-scalable asset-specific reference. The comparison intentionally favours the reference: the asset-specific autoencoder is trained and calibrated independently using the complete healthy dataset of each operating condition, whereas the proposed methodology performs no field-model training and estimates only a local operational radius through convergence-based commissioning. Across the four CWRU conditions, commissioning required only 49–62 nominal observations.

TABLE 4 Ideal but non-scalable asset-specific deployment versus the proposed fleet-scalable deployment. Values are mean ± standard deviation across the four CWRU operating conditions. Inner-race DR was 100.00 ± 0.00 for all three strategies and is omitted. The last column states what each strategy requires from every deployed asset; the shared field-oracle strategy uses the complete healthy dataset only to estimate the local radius and does not retrain the shared representation.

Deployment strategy	FAR (%)	Ball DR (%)	Outer DR (%)	Required per asset
Asset-specific oracle	2.29 ± 0.12	100.00 ± 0.00	100.00 ± 0.00	Model + all healthy data
Shared AE, oracle radius	2.29 ± 0.12	99.45 ± 0.96	99.96 ± 0.07	All healthy data
Proposed: online radius	2.60 ± 1.18	99.57 ± 0.71	100.00 ± 0.00	49–62 observations

Replacing four independently trained autoencoders with one simulator-trained shared representation produced no observed loss in inner-race detection and no practically relevant loss in outer-race or ball-fault detection. Relative to the asset-specific reference, average ball-fault detection decreased by only 0.43 percentage points, while the mean healthy false-alarm rate increased from 2.29% to 2.60%.

The more direct comparison is with the shared representation using a complete-data field-oracle radius. In that comparison, the online procedure achieved slightly higher average ball- and outer-race detection, together with a 0.31 percentage-point increase in healthy FAR. Thus, nearly all of the diagnostic performance available from complete retrospective calibration was recovered through an automatically terminated commissioning phase requiring at most 62 observations.

The principal gain is therefore organizational and computational rather than algorithmic. The asset-specific reference requires an independently fitted preprocessing transformation, representation model, validation procedure, and complete-data calibration for every operating condition. The proposed architecture retains one centrally engineered representation and requires each deployed asset to identify only one scalar local operational radius.

Table 5 examines this local identification directly by comparing the converged online radius with the empirical p98 radius obtained retrospectively from every available healthy observation.

TABLE 5 Agreement between the converged online operational radius and the complete-data field-oracle p98 reference. Relative difference is computed as $100|\tau_{\text{online}} - \tau_{\text{oracle}}|/\tau_{\text{oracle}}$. Post-calibration FAR is evaluated only on healthy windows occurring after the declared convergence point.

RPM	τ_{oracle}	τ_{online}	Relative diff. (%)	Cal. windows	Post-cal. FAR (%)
1730	0.3578	0.3520	1.64	49	3.94
1750	0.3434	0.3440	0.19	62	1.75
1772	0.3512	0.3550	1.06	58	0.87
1797	0.3886	0.3869	0.44	49	3.90
Mean	–	–	0.83	54.5 ± 6.6	2.62

The converged online radii differ from their complete-data oracle references by only 0.19%–1.64%, with a mean relative difference of 0.83%. Moreover, the post-calibration healthy FAR remains close to the nominal 2% operating point, averaging 2.62% across the four conditions. The local commissioning procedure therefore does not merely preserve fault-detection performance; it recovers almost the same asset-specific acceptance boundary that would have been obtained with retrospective access to the complete healthy dataset.

These results support the central deployment claim of the paper. A single simulator-trained nominal representation, combined with convergence-based local commissioning, approaches the diagnostic operating point of independently engineered asset-specific models while eliminating asset-specific representation learning and complete-data calibration.

Operational-radius convergence

Figure 1 shows the sequential estimation of the local operational radius under the fixed shared representation. During commissioning, neither the absolute simulator threshold nor a standardized simulator residual scale is transferred. The only fleet-wide calibration quantity is the target nominal coverage, $1 - \alpha = 0.98$.

For operating condition k, the operational radius after observing t commissioning residuals is estimated directly as

$$\tau_k(t) = \widehat{Q}_{k,t}(0.98), \tag{3}$$

where $\widehat{Q}_{k,t}(0.98)$ denotes the empirical 98th percentile of the local reconstruction residuals observed up to commissioning instant t. The shared representation remains unchanged throughout this process; only the local operational radius is identified.

$$C_k(t) = \frac{|\tau_k(t) - \tau_k(t-L)|}{\max\{|\tau_k(t-L)|, \epsilon\}}, \tag{4}$$

where L is the look-back interval and $\epsilon > 0$ is a numerical stabiliser preventing division by zero.

Convergence is evaluated only after the minimum calibration size $N_{\min} = 40$. Using the relative-variation criterion defined in Eq. (4), commissioning terminates when $C_k(t) < \varepsilon_C$ for m consecutive updates.

In all reported experiments, $L = 10$, $\varepsilon_C = 0.01$, and $m = 10$. If convergence is not detected before $N_{\max} = 200$, the asset is assigned a calibration-failure status and is not released into active monitoring.

For retrospective comparison, the complete-data field-oracle radius is

$$\tau_k^{\text{oracle}} = \widehat{Q}_{k,\text{all}}(0.98), \tag{5}$$

where the quantile is computed using every available healthy field window for operating condition k. This quantity is unavailable during commissioning and is used only to assess agreement with the operating point that would be obtained under complete nominal knowledge.

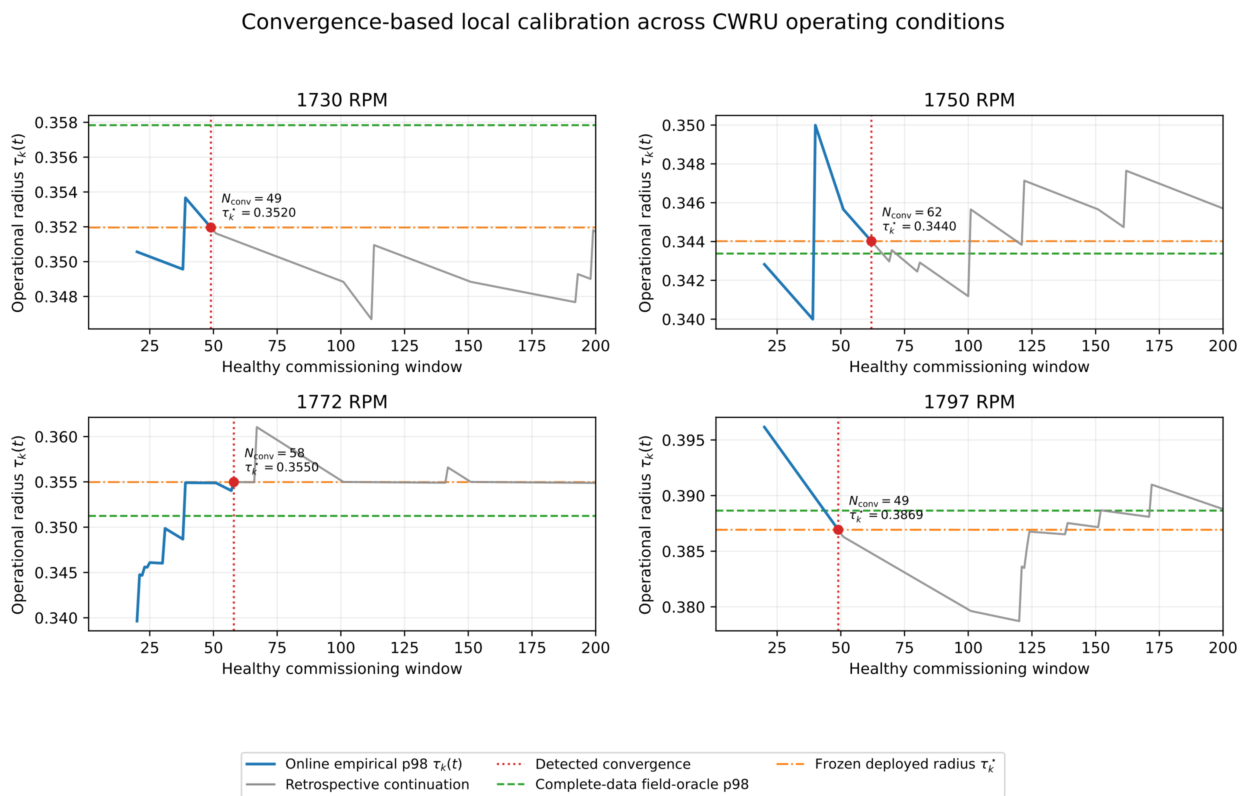

FIGURE 1 Sequential estimation of the empirical p98 operational radius during CWRU commissioning. For each operating condition, the blue segment shows the online estimate defined in Eq. (3), $\boldsymbol{\tau_k(t) = \widehat{Q}_{k,t}(0.98)}$**, until the convergence criterion is satisfied. The red dotted line marks the declared convergence point, at which the operational radius is frozen and active monitoring begins. The gray segment is a retrospective continuation obtained by incorporating additional observations known from the benchmark to be healthy; it is shown only to characterize the subsequent behaviour of the estimator and is not used to modify the deployed operational radius. The horizontal dashed line indicates the complete-data field-oracle reference defined in Eq.** (5)**.**

The retrospective trajectories show that an empirical upper quantile may change again after an interval of apparent stability when additional high-residual nominal observations arrive. This behaviour is expected because, during short commissioning periods, the empirical p98 estimate is determined by relatively few upper-tail observations. The gray trajectories therefore do not represent continued adaptation of the deployed detector; they show how the estimator would have evolved had commissioning remained active.

The operationally relevant result is the agreement between the frozen online radius and the complete-data oracle at the declared stopping point. Despite using only 49–62 commissioning observations, the online procedure recovers a local operational radius within 2% of the value obtained retrospectively from all available healthy observations. Post-calibration healthy false-alarm rates range from 0.87% to 3.94%, remaining close to the nominal 2% operating point.

The convergence criterion should not be interpreted as proof that the population 98th percentile has been recovered exactly. It is an operational stopping rule indicating that the estimated operational radius has remained sufficiently stable under the selected configuration $(N_{\min}, N_{\max}, L, \varepsilon_C, m)$. These parameters define a configurable trade-off between commissioning duration and sensitivity to temporary upper-tail plateaus.

An asset that does not converge within $N_{\max}$ is not released into active monitoring and is assigned a calibration-failure status. Such a failure may indicate insufficient commissioning data, non-stationary nominal operation, or initial behaviour inconsistent with the assumed nominal state. Distinguishing among these causes lies outside the present evaluation.

Robustness to local initialization

A practical deployment question is whether convergence-based local calibration depends critically on the particular nominal observations encountered during commissioning. To evaluate this sensitivity, the shared simulator-trained representation, preprocessing transformation, target coverage, and convergence parameters were kept fixed throughout the experiment. Only the nominal observations available during commissioning, together with their arrival order, were varied.

For each CWRU operating condition, five independent commissioning sequences were generated by randomly permuting the available healthy observations. Calibration was allowed to start after $N_{\min} = 40$ observations and continued until the empirical p98 operational radius satisfied the fixed convergence criterion or the maximum commissioning budget $N_{\max} = 200$ was exhausted. Once convergence was declared, the estimated operational radius was frozen and subsequently used for fault detection. Healthy false-alarm rate was evaluated only on healthy observations excluded from calibration, while all fault classes were evaluated using the same shared representation without retraining.

Table 6 summarizes the results obtained across the five independent commissioning sequences.

All 20 commissioning runs satisfied the convergence criterion before reaching the maximum calibration budget, yielding a calibration success rate of 100%. Across all operating conditions, the estimated operational radius remained within $2.08 \pm 0.71\%$ of the complete-data oracle, while the fleet-averaged post-calibration healthy false-alarm rate was $3.59 \pm 1.59\%$. Both quantities are computed in two stages: for each of the five commissioning seeds, the metric is first averaged across the four CWRU operating conditions, yielding five seed-level values; the reported mean and standard deviation are then taken across these five values. Fault-detection performance remained essentially unchanged throughout all commissioning realizations, indicating that the complete deployment procedure is robust to realistic variability in the local initialization sample and its arrival order.

TABLE 6 Robustness of convergence-based local calibration to commissioning sample variability. The shared simulator-trained representation and all calibration parameters remained fixed; only the healthy commissioning observations and their arrival order were varied. Values are reported as mean ± standard deviation over five independent commissioning realizations. Inner-race DR was 100.00 ± 0.00 under every condition and is omitted. FAR is evaluated only on healthy observations excluded from calibration.

RPM	τ_{online}	Oracle error (%)	Post-cal. FAR (%)	Ball DR (%)	Outer DR (%)	Cal. windows
1730	0.3486 ± 0.0023	2.59 ± 0.63	5.38 ± 1.39	99.95 ± 0.11	100.00 ± 0.00	55.6 ± 8.2
1750	0.3451 ± 0.0111	2.53 ± 1.65	4.23 ± 5.63	99.90 ± 0.14	100.00 ± 0.00	57.4 ± 7.9
1772	0.3511 ± 0.0086	1.82 ± 1.37	2.54 ± 2.18	100.00 ± 0.00	99.97 ± 0.06	54.4 ± 5.9
1797	0.3915 ± 0.0065	1.40 ± 1.02	2.21 ± 2.23	97.98 ± 0.71	99.97 ± 0.06	64.4 ± 26.0

These experiments evaluate the robustness of the complete commissioning procedure rather than the variability of a fixed-size percentile estimate. The representation, preprocessing pipeline, target coverage, convergence criterion, and all deployment parameters remained unchanged. Only the local healthy observations available during commissioning and their arrival order were modified.

Some variability is expected because the empirical p98 operational radius is estimated from a relatively small number of upper-tail observations. Consequently, different commissioning samples naturally produce slightly different estimates of the local operational radius. These differences translate into modest variations in the healthy false-alarm rate while leaving the resulting diagnostic operating point essentially unchanged.

The largest healthy false-alarm rate observed during the repeated commissioning experiments was 13.8%. However, this isolated realization did not alter the deployment outcome: convergence was still achieved, the estimated operational radius remained within 4% of the oracle estimate, and fault-detection performance remained essentially unchanged. The observed spread reflects finite-sample uncertainty in estimating an extreme empirical quantile rather than instability of the deployment methodology. Its direction, however, is systematic: post-calibration rates lie above the nominal 2% operating point in every condition. This is consistent with the stopping rule, which terminates when the upper-tail estimate has been locally stable and therefore tends to stop at instants of quiet tail behaviour, biasing τ slightly downward. Linear interpolation in the empirical quantile, which returns a value below the corresponding order statistic, acts in the same direction.

Together with the oracle-agreement results of Table 5, these experiments establish that convergence-based commissioning reliably recovers the operational neighbourhood of the shared nominal manifold under realistic initialization variability. The oracle-agreement experiment evaluates chronological calibration against the complete-data reference, whereas the present experiment demonstrates that essentially the same neighbourhood is recovered under different commissioning realizations without modifying the shared representation.

The independent NASA IMS validation presented in Section 5.7 extends this conclusion beyond controlled operating conditions. There, independently commissioned physical assets converged to distinct local operational radii while subsequently exhibiting persistent departures from their respective nominal neighbourhoods during naturally evolving degradation. Taken together, the CWRU and NASA experiments support the central deployment hypothesis of this work: the transferable object is the shared nominal representation, whereas each asset identifies only its own local operational radius during commissioning.

Multi-unit run-to-failure deployment on NASA IMS

The CWRU experiments demonstrate simulation-to-field transfer under controlled operating conditions. The NASA IMS experiments evaluate the complete deployment methodology under realistic fleet conditions, where independently operating physical assets undergo naturally evolving degradation over their remaining service life.

Each NASA bearing is treated as an independent monitored asset. The simulator-trained autoencoder and preprocessing pipeline are transferred unchanged, and no NASA observations are used for representation learning, retraining, or parameter adaptation.

Table 7 summarizes the deployment results. Operational-radius calibration converged successfully for all four bearings after only 49–57 healthy recordings. Although the estimated operational radii differ substantially across bearings, every asset uses exactly the same shared representation, the same target nominal coverage (p98), and the same convergence criterion without bearing-specific retraining. The only asset-specific operation is the estimation of the local operational radius from the early-life recordings selected for commissioning.

Following commissioning, all four bearings subsequently departed persistently from their estimated nominal neighbourhood. The learned operational radius remains predictive throughout the subsequent degradation trajectory, with a mean post-departure exceedance rate of 94.6% and lead times ranging from 14.5 h to 74.0 h before the documented end of the experiment.

TABLE 7 NASA IMS deployment results using one shared simulator-trained representation and convergence-based local operational-radius estimation.

Bearing	Cal. windows	τ_k	Relative error (%)	Post-departure exceedance (%)	Lead time (h)
1	49	0.010680	0.10	99.12	74.00
2	57	0.014950	0.35	86.42	38.83
3	49	0.018930	2.87	95.88	14.50
4	49	0.006672	0.18	97.02	54.33
Mean	51	–	0.87	94.61	45.42

Bearing 3 exhibits the largest relative deviation from its oracle radius (2.87%) among the four monitored units. This is consistent with its higher sensitivity to commissioning-sample variability, as quantified in Section 5.7.1.

These results demonstrate that the transferable engineering artefact is the shared nominal representation rather than an asset-specific detector. The same representation identifies substantially different operational radii for different physical assets while requiring neither representation retraining nor threshold engineering. Once estimated during commissioning, each operational radius continues to separate nominal from degrading behaviour throughout the remaining lifetime of the asset.

Stability of local operational-radius estimation

A practical deployment methodology must be robust to the particular early-life observations encountered during commissioning and to their arrival order. To evaluate this property, the shared representation was kept fixed while only the commissioning observations and their order were varied.

For each NASA bearing, five independent commissioning realizations were generated by sampling and permuting the first 200 early-life recordings. No retraining was performed; only the observed nominal residuals differed between runs.

Across twenty independent commissioning experiments (four bearings and five random initialization samples), calibration converged successfully in every case. Table 8 summarizes the resulting variability.

TABLE 8 Stability of local operational-radius estimation under repeated NASA commissioning experiments using one fixed shared representation. Calibration converged in all twenty realizations (success rate 100%).

Bearing	τ_{mean}	τ_{std}	Relative error (%)	Healthy FAR (%)	Cal. windows
1	0.010590	0.000074	0.73 ± 0.70	5.84 ± 3.79	49.2 ± 0.4
2	0.014873	0.000084	0.36 ± 0.44	3.71 ± 3.33	49.0 ± 0.0
3	0.018151	0.000383	1.58 ± 1.88	4.93 ± 4.22	52.2 ± 7.2
4	0.006640	0.000033	0.66 ± 0.49	5.45 ± 2.39	49.6 ± 1.3
Mean	–	–	0.83	4.98	50.0

The observed spread reflects finite-sample uncertainty in estimating an extreme empirical quantile rather than instability of the deployment methodology. Its direction, however, is systematic: post-calibration rates lie above the nominal 2% operating point in every condition. This is consistent with the stopping rule, which terminates when the upper-tail estimate has been locally stable and therefore tends to stop at instants of quiet tail behaviour, biasing τ slightly downward. Linear interpolation in the empirical quantile, which returns a value below the corresponding order statistic, acts in the same direction.

These experiments demonstrate that the operational radius is simultaneously asset-specific and statistically stable. Different physical units identify different local neighbourhoods of the same shared nominal representation, whereas repeated commissioning of the same unit produces only minor variation in the estimated radius. The transferable engineering artefact is therefore the shared nominal representation itself, while commissioning consists only of identifying the local operational neighbourhood associated with each physical asset.

Edge deployment implications

The shared model contains 847,601 trainable parameters (3.39 MB as raw float32, 10.24 MB serialized). Median CPU inference latency over a 1,200-sample window is 6.26 ms (p95: 6.62 ms), well within the 100 ms acquisition interval. Per-asset state is limited to a 200-sample residual buffer of 800 bytes, giving under 1 kB of calibration state; no per-asset model training and no fault labels are required after installation.

The only fleet-wide artifact is the shared nominal representation, trained once and distributed to every operational unit.

Limitations

Several limitations qualify the scope of the present results and should guide their interpretation.

Nominal initialization assumption. The procedure assumes the cold-start observations are representative of healthy operation. If an asset were already degraded at installation, $\tau_k^\star$ could absorb part of that condition into the nominal baseline; a consistency check against the canonical nominal manifold is a natural extension and is left as future work.

Short-commissioning sensitivity. Because the operational radius is an empirical 98th percentile estimated from a limited number of upper-tail observations, it remains sensitive to the particular nominal observations encountered before convergence. This effect was visible in the repeated-commissioning experiment, where one 1750 RPM realization produced a 13.8% post-calibration FAR despite remaining within 4% of the complete-data oracle. Longer commissioning, more restrictive convergence parameters, or robust quantile estimators may reduce this variability at the cost of later deployment.

Stationarity of nominal operation. The convergence criterion assumes an approximately stationary healthy residual distribution during cold start. The retrospective trajectories in Figure 1 show modest post-calibration drift; more variable startup conditions were not evaluated and may benefit from a longer or adaptively triggered calibration window.

Validation domain. The validation uses bearing datasets (CWRU, NASA IMS) rather than instrumented transportation assets. While the deployment principle is architecture- and modality-agnostic by construction (Section 3), its extension to other components remains to be confirmed empirically.

Future Work

A natural extension relaxes the assumption that assets begin in nominal condition. Because the methodology separates the shared representation from the local radius, future work will investigate validating the initialization phase itself against the canonical nominal manifold, to distinguish genuinely healthy startup from incipient degradation before it is absorbed into the baseline. A second direction concerns progressive degradation: whether the joint evolution of reconstruction residuals and operational-radius dynamics carries predictive information, connecting this methodology with physics-informed health indicators and remaining-useful-life estimation.

Finally, the mathematical interpretation in (Di Santi 2026a) suggests formalizing the operational radius as an asset-specific neighborhood around the transferable canonical representation, characterizing its invariance properties and the conditions under which local calibration preserves the underlying quotient-space structure.

Finally, the proposed methodology yields detection rather than diagnosis. Where a kinematic model is available, envelope analysis attributes a departure to a specific fault surface; the operational radius reports only that an observation lies outside the nominal neighbourhood.

CONCLUSION

We presented a day-one deployment methodology for nominal condition monitoring across heterogeneous asset fleets, validated on public rolling-element bearing datasets. Within each compatible asset family, one nominal representation is trained once on simulated healthy observations and transferred unchanged to all deployed units. Each operational unit then performs only a short unlabeled warm-up to estimate its own operational radius, without modifying or retraining the shared representation.

The experimental results show that a simulator-trained nominal representation transfers successfully to field deployment, whereas an absolute decision threshold does not. A shared target nominal coverage, realized locally through the empirical residual quantile estimated during initialization, provides an effective bridge between the shared representation and field deployment. This deployment strategy preserves detection performance close to that of asset-specific models trained under ideal asset-specific calibration conditions while eliminating per-asset representation learning.

The transferable object is therefore not an absolute anomaly threshold, nor an asset-specific representation, but a shared canonical nominal representation together with a lightweight local calibration mechanism. The central outcome of this work is consequently not a new anomaly detector, but a scalable deployment principle: learn the canonical nominal representation once, distribute it across the fleet, and let each physical asset identify only its own operational radius during installation.

This separation between representation learning and deployment calibration provides a practical foundation for large-scale edge condition monitoring, where an OEM can centrally produce and maintain the shared representation while each deployed asset performs its own lightweight local calibration automatically, without operator intervention or manual threshold tuning.

Conventional CBM deployments typically require asset-specific engineering of models and alarm thresholds. The present results suggest that this engineering effort can instead be replaced by a standardized commissioning procedure in which every asset estimates its own operational radius while reusing a common nominal representation. The representation is shared. The operational radius is individual.

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